

Colorization Network Watermarking in the CIE-Lab Domain

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Abstract—A possible solution to protect the copyright of generative models is to watermark the models so that any image generated by the models contain an invisible watermark, whose presence can be checked at a later stage for ownership verification or to trace back the image to the generator which produced it. In general, a Generative Adversarial Network (GAN) or a diffusion model can be watermarked by applying a frozen pretrained watermark decoder on top of the generator, and adding the watermark decoding loss term to the generator loss. In this paper, we propose a method to watermark image colorization models, that is models whose goal is to introduce plausible colors in grey-level images. The particular color domain wherein colorization models operate requires that the watermark is embedded in the image components that are actually affected by the colorization, avoiding to embed the watermark in the luminance channel, which is usually left unchanged by the colorization process. In particular, we show that the domain the watermark decoder is trained on impacts the performance of the network and better performance in terms of watermark accuracy and robustness can be achieved by training the decoder to extract the watermark bits from the chrominance components in the CIE-Lab space and use the decoder trained in this way to watermark the GAN model.

Index Terms—Image colorization, DNN watermarking, watermarking of generative models, watermarking in the CIE-Lab domain

I. INTRODUCTION

Watermarking has been introduced as a solution to protect the Intellectual Property Rights (IPR) of Deep neural Networks (DNNs) [1]–[3]. Early works focused on the watermarking of Convolutional Neural Networks (CNNs), with particular attention to CNN-based classifiers [1], [4]. Due to the impressive progress in generative technologies, the focus has recently shifted to the watermarking of image generative models [5], [6] and large language models [7]. In this case, due to the large entropic content of the output of the models, it is possible to ask that the watermark be extracted from every output of the watermarked model [6], [8]. In the case of image generative models, this means that the watermark is read from every image produced by the network, a situation often referred to as joint DNN/image watermarking. In most cases, watermark embedding is achieved by separately training a watermark decoder, and then train the generative model making sure that the decoder reads the desired bits from the generated images [6], [9].

In this work we focus on the watermarking of new kind of models, namely colorization models, whose goal is to colorize an image starting from its grey-level version [10]. While image colorization can be seen as a particular type of image processing task, it has some peculiarities that require the development of an ad-hoc watermarking solution. Specifically, a good colorization network is supposed to leave the luminance component of the input image unchanged and operate only on the chrominance components. Therefore, it is necessary that the watermark does not modify the image luminance and is embedded only in the chrominance components of the image. This makes the adoption of off-the-shelf solutions, where the watermark decoder extracts the watermark by analysing the watermarked image in the RGB domain, unsuitable. In this paper, we propose a method expressly thought to embed the watermark in the chrominance domain, in particular in the a and b components of the CIE-Lab color space. To do so, we first train a watermark encoder/decoder

pair by forcing them to work in the ab domain. Then we freeze the watermark decoder, and use it to simultaneously train and watermark the colorization model. During training, we pay particular attention to make sure that the watermark can be retrieved also from a processed version of the colorized images. Inspired by [6], [9], in our method such robustness is achieved by introducing a noise layer, applying standard processing operations, like JPEG compression and Gaussian noise addition, before the watermark decoder. The noise layer is applied during the training of both the watermark decoder and the to-be-watermarked colorization network.

We evaluated the effectiveness of our approach by using it to watermark a GAN-based colorization model and comparing the performance we got with those achieved by a state-of-the-art system working in the RGB domain. The results of our experiments confirm the validity of our intuition, showing that embedding the watermark in the ab domain allows to improve the accuracy and robustness of the watermark, with no significant impact on the quality of the images generated by the model. Our experiments also reveal that our watermarking scheme working in the Lab domain achieves noticeable robustness against post-processing, and the watermark can be retrieved from images processed with various kinds of operators, also when the processing is applied with a pretty large strength. Notably, robustness is also achieved against simulated print and scan (P&S) [11]. Eventually, the watermark shows a good robustness also against model simplification (pruning).

II. RELATED WORK

A. Watermarking of generative models

The first work proposing a method for instructing GANs to add an invisible watermark to the images they produce is [8]. In this method, a given invisible watermark is first embedded into the training data by exploiting a pre-trained image watermarking network, then the watermarked data is used to train the GAN. In this way, the watermark present in the training data can also be retrieved from the images generated by the GAN. In [6], a method is proposed to watermark GANs by enforcing the presence of a specific watermark within the generated images, exploiting a pre-trained image watermarking decoder to impose an additional watermarking loss term during training. The work in [6] has been extended to diffusion models in [9]. In [12] a similar approach is adopted to watermark networks accomplishing a wide variety of image processing tasks. In this case, the network accomplishing the original task and the watermark-extraction network are trained together. All the above schemes work in the RGB domain, implicitly assuming that the generative task is about learning a mapping from input and output RGB images. When this is not the case, like in the case of colorization networks, the domain where the watermark encoder/decoder pair works may have a significant impact on the effectiveness of the watermarking process.

B. Image colorization

Image colorization is an active research area with primary application to the restoration and enhancement of old and degraded

images/videos. Colorizing an image involves enriching a grayscale image with suitable colors fitting the content of the scene [10]. Image colorization is a problem which has multiple plausible solutions, since for every grayscale image multiple plausible color versions can be produced. With the advancement of neural networks, the most recent methods for colorization are based on DNNs. Despite some scattered approaches which learn the mapping from the gray-scale images directly to their RGB counterpart [13], i.e. working in the RGB domain, the majority of the methods address the colorization problem in the CIE-*Lab* domain (or any other suitable color domain separating the chrominance and luminance components), by generating only the *a* and *b* channels of the image (representing the green-red and the blue-yellow opponent pair of the human vision model), and leaving the luminance channel *L* equal to that of the original gray-scale image. In [14], a CNN is trained to map from a gray-scale input to the *ab* domain. To reduce the tendency to produce low-saturated colors, a custom prediction loss is used, specifically tailored to the colorization problem. Conditional GANs (cGANs) have also been adopted for colorization [15]. The addition of an L1 regularization term in the loss mitigates the tendency of cGANs to generate too colorful images, thus producing more realistic renderings and color distributions. In [16], an approach is proposed to colorize grayscale images using GANs and semantic information, where the generative network infers the *a* and *b* components of a given gray-scale image conditioned on semantic clues. Some works have shown that image-to-image diffusion models can also work well for colorization without task-specific specialization [17]. More recent approaches adopt network architectures where the colorization is guided by means of additional inputs, like textual descriptions. Examples of language-guided colorization models can be found in [18]–[21], where textual inputs are used to describe how certain objects should be colored.

III. SUPERVISED WATERMARKING IN THE CIE-LAB CHROMINANCE DOMAIN

Figure 1 illustrates the overall pipeline of the proposed method for the watermarking of a colorization network in the CIE-*Lab* domain. The goal is to watermark the network G in such a way that a given watermark message $w \in \{0, 1\}^l$ of length l is embedded in all the images generated by G . The watermark is retrieved by applying a pretrained frozen decoding network D_w capable of embedding the target payload l . The watermark decoding network D_w is trained during a preliminary phase, where a pair of watermark embedding/decoding networks is trained. The decoder is then used to guide the optimization of the colorization network. Let x denote the input image (RGB domain) and x^{Lab} denote the image in the *Lab* domain. We indicate with x^L and x^{ab} the luminance (*L*) channel and the chrominance (*ab*) channels of x . We stress that to embed a different sequence of bits (say w'), there's no need to train a new watermark encoder/decoder pair (assuming that the payload does not change) and only the GAN must be retrained.

Image watermarking networks. The watermarking encoder/decoder pair works directly in the *ab* domain, discarding the luminance component. By training the watermarking networks in this way, the decoder is instructed to *read* the watermark from the chrominance channels of the image. In fact, these are the only components where the watermark can be embedded by the colorization network.

Colorization network watermarking. When the colorization network is trained, the generator G is instructed to predict the *ab* chrominance components starting from the grayscale image (*L* channel), in such a way to fool a discriminator D and produce a plausible colorized image. The generated image is also input to D_w

for watermark extraction, hence the colorization and the watermark embedding tasks are learned simultaneously. Note that, although in principle the discriminator could just look at the *ab* channels, the luminance information is also relevant to judge the plausibility of the colorization. Specifically, the discriminator is optimized by minimizing the following loss (binary cross-entropy):

$$L_D = \sum_{i=1}^n \text{BCE}(D(x_i^{Lab}), 1) + \sum_{i=1}^n \text{BCE}(D([x_i^L; G(x_i^L)]), 0), \quad (1)$$

where $\text{BCE}(x, y) = -y \log x - (1 - y) \log(1 - x)$, n is the number of training images, and the generator is trained to minimize a combined loss $L_G^w = L_G + \gamma L_w$. L_G is a conventional generator loss:

$$L_G = - \sum_{i=1}^n \text{BCE}(D([x_i^L; G(x_i^L)]), 1) + \lambda \sum_{i=1}^n \|x_i^{ab} - G(x_i^L)\|_1, \quad (2)$$

with $\lambda > 0$, ensuring that the distance between the colorized image and the ground truth is small (right term) and that the discriminator is not able to distinguish the original and colorized images (left term). $L_w = \text{BCE}(D_w(G(x_i^L)), w)$ (averaged on all the watermark bits) is the watermarking loss, whose role is to make sure that the watermark decoder reads the correct bits from the colorized images. Finally, γ is a parameter balancing the importance of the watermark decoding error and the quality of the colorized images.

Image processing layer. A noise layer (image processing layer) is included to improve the robustness of the embedded watermark. By asking the decoder to retrieve the watermark also from a processed version of the images, the encoder is forced to embed the watermark in components of the images that are modified less by the processing. In the noise layer, the image is first converted from the *Lab* to the RGB color space, since all most common processing operators are applied in the RGB domain. After processing, the image is converted back to the *Lab* space, and the *ab* components are fed to the decoder for watermark extraction. We observe that the mapping from RGB to *Lab* is an easily invertible transformation, allowing a straightforward application of backpropagation.

IV. EXPERIMENTAL SETTING

We validated our scheme by running experiments on facial images from the CelebA-HQ dataset [22]. A total of 30K images is considered, split as follows: 21000 images are used for training, 5999 and 3001 for testing and validation respectively. We considered a payload of 100 and 200 bits for the watermark w , that is, $l = 100$ and 200.

A. Network architectures

We used the original PatchGAN discriminator in the Pix2pix network [15] to implement the discriminator D (note that, in principle, any other discriminator could be used), while we resorted to the colorization network in [14] to build G . Similarly to [6], we embedded the watermark by fine-tuning a pre-trained colorization network, which reduces the computational burden and allows the protection of an already trained colorization networks with little cost. We also trained a version of the watermarked colorization network with D_w working in the RGB domain as a baseline method to validate our hypothesis that embedding and reading the watermark in the *ab* domain allows better performance. As watermarking encoder-decoder network, we adopted the StegaStamp network [23]. We modified the input layer of the StegaStamp encoder to accommodate the 2 channel input x^{ab} and added a layer on top of it performing the RGB to *Lab* conversion. We also trained a version of StegaStamp working in the RGB domain and used it to build the baseline colorization network. The input size of the images was set to 256×256 . For training

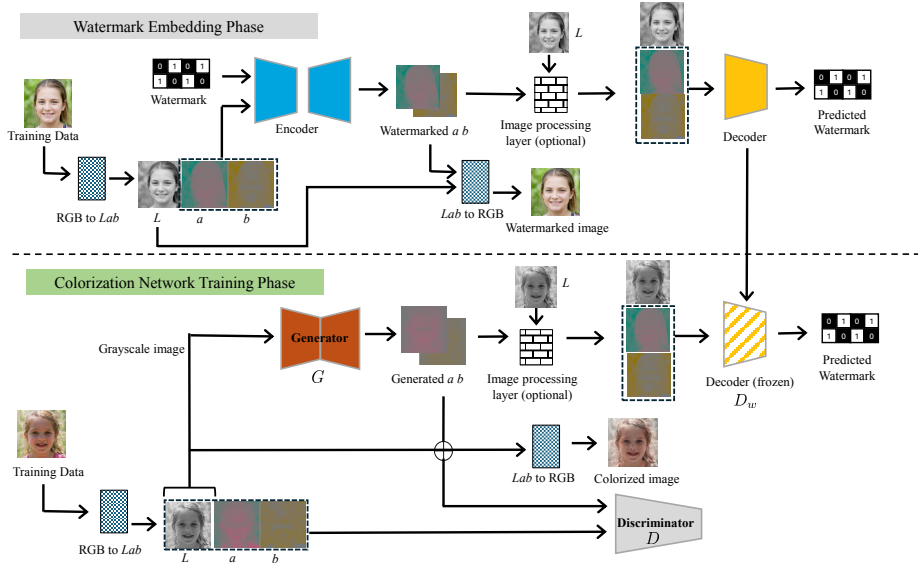


Fig. 1. Pipeline of supervised colorization watermarking. In the first stage (upper image), a watermarking encoder/decoder pair working in the ab domain is trained. The pre-trained decoder is frozen and used during to train the image colorization GAN (bottom part). The noise layer (image processing layer) is included during the training of both the watermark encoder/decoder and the colorization network. As most of the processings are applied in the RGB domain, Lab to RGB conversion is applied inside the noise layer prior to the application of the processing; the image is then converted back to Lab after processing (these conversions are not indicated explicitly in the figure).

with the noise layer, we considered JPEG compression (namely, the differentiable approximation in [24]) with QF in [45:85], and Gaussian noise addition (with standard deviation in $[0, 0.2]$).

B. Training settings

We trained the ab version of StegaStamp for 15 epochs, with the first epochs of warm up, where the network learns only how to retrieve the watermark from the images, and the last ones focus on both watermark retrieval and image reconstruction, using the same setting described in [23]. For the colorization GAN, we adopted the same generator used in [14], while for the discriminator we used a PatchGAN network. The GAN was first trained to learn the colorization task ($\gamma = 0$), then we fine-tuned it by adding the watermarking loss term. With regard to the parameters λ and γ balancing the various terms of the training loss, we let $\gamma = 0.05^1$ and $\lambda = 10$. We fine-tuned the GAN for 6 epochs with batch size 32, using Adam optimizer and learning rate 10^{-7} for the discriminator (10^{-4} for the generator). The RGB baseline StegaStamp and watermarked GAN models were trained using the same setting.

¹This value was found using a grid search in the $[0.01:0.5]$ range for $l = 100$, then we adopted the same value for $l = 200$. We found experimentally that using a larger γ introduces visible artefacts in the watermarked images.

TABLE I
BIT ACCURACY AND FID VALUES FOR VARIOUS WATERMARKED MODELS.

Model	l (bits)	Noise layer	Bit Acc	FID
Baseline - No wat	-	-	-	7.13
Baseline - RGB	100	yes	82.2%	11.40
	200	yes	74.9%	10.20
Prop - Lab	100	no	97.5%	9.02
		yes	99.5%	11.12
	200	no	99.1%	7.63
		yes	93.6%	9.72

V. RESULTS

A. Colorization quality and watermark accuracy

To evaluate the quality of the generated images, we used the Frechet inception distance (FID), obtained from statistical quantities measured from the two distributions of real and generated images. In fact, although in this case an original counterpart of the generated images is available, the problem of colorization of an image has multiple plausible solutions, hence the similarity with respect to the original color image (measured via the PSNR or SSIM) does not provide a good indication of the goodness of the colorization. Table I reports the bit accuracy and the FID obtained with our watermarking scheme and the baseline in the various settings. In all the cases, when working in the Lab domain the decoder can extract the watermark with almost perfect accuracy. As expected, slightly worse FID values are obtained with the inclusion of the noise layer. This is the price to pay to get a more robust watermark. The FID values in the case of watermarked networks are aligned with the value obtained with the non-watermarked model, confirming that the presence of the watermark does not affect much the quality of the generated images and the colorization capability of the network. A significant improvement of the watermark accuracy for similar (in fact slightly better) values of FID can be observed with respect to the baseline models embedding the watermark in the RGB domain. Some examples of colorized images obtained with $l = 100$ and 200 are reported in Figure 2. From the figure, we see that the watermark is embedded with almost no impact on image quality. The colorization capabilities remain good for both the proposed Lab domain networks and the baseline networks working in the RGB domain.

B. Robustness performance

To start with, we evaluated the importance of the noise layer to obtain a robust watermark. To do so, we measured the robustness of the watermark against JPEG compression and Gaussian noise

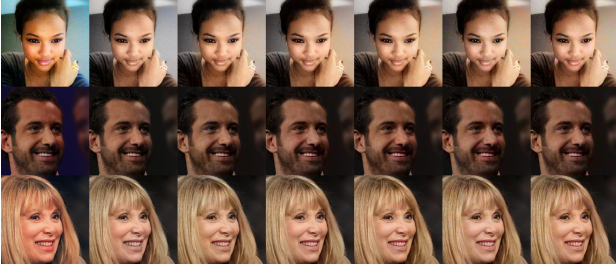


Fig. 2. From left to right, example of: (a) ground truth color images, and images generated by a colorization model with: (b) no watermark, (c) *Lab* watermark with $l = 100$ and no noise layer, (d) *Lab* watermark with $l = 100$ and noise layer, (e) baseline RGB watermark with $l = 100$ (with noise layer), (f) *Lab* watermark with $l = 200$ and noise layer, (g) baseline RGB watermark with $l = 200$ (with noise layer).

addition. The experiments we run revealed that without using the noise layer the accuracy of the watermark drops to less than 70% even with the addition of mild Gaussian noise with standard deviation equal to 0.05, and to less than 80% in the presence of JPEG compression with QF = 80. For this reason in the following we report only the results obtained by using the noise layer during training.

The results reporting the robustness of the various models trained with the noise layer against image processing attacks are shown in Figure 3 and Table II. In Figure 3, we show the robustness against JPEG compression and noise addition, which are the processing operators included in the noise layer. As it can be seen, the inclusion of the noise layer allows to significantly improve the robustness against processings. With regard to the comparison between RGB and *Lab* watermarking, by focusing on the case of JPEG compression, we see that working in the *ab* domain allows to improve the robustness of almost 20% in the 100 bits case. A similar gain is observed for the 200 bits case in the high QF regime. It is also noticeable that in the 100 bits case, the accuracy remains very close to 100% when $QF \geq 70$, and well above 90% even when QF approaches 40. A similar trend is observed for the case of Gaussian noise addition. Table II reports the results of the robustness tests against image processing operators that were not included in the noise layer. In particular, we tested the robustness of the watermark against the following operators: blurring, median filtering (5×5 and 7×7), and Print and Scan (P&S) simulation implemented as in [11]. We also included two color and brightness transformations which may be particularly relevant for a system working on image colors. In particular, we considered brightness/contrast adjustment and hue/value/saturation adjustment (referred to as HSV Transf. in the table). The former is defined by two parameters α and β , where α is a multiplier that controls the contrast of the image and β is an additive factor that adjusts the brightness of the image. In our experiments we set $\alpha = 0.85$ and $\beta = 15$. The latter shifts the hue component of the image, and scale the saturation and brightness of the image. In our experiments we set the shift and scale values to 3, 1.15 and 0.8 respectively. Even in this case, the accuracy of the watermark introduced in the *ab* component is consistently better than that obtained with the models watermarked in the RGB domain. The gain of the *Lab* models is always larger than 10% (often much more than that) and is particularly relevant for the P&S simulations, where the models watermarked in the RGB domain achieve an accuracy close to a random guess. A possible explanation is that the printing and scan process affects the textures more than the image colors.

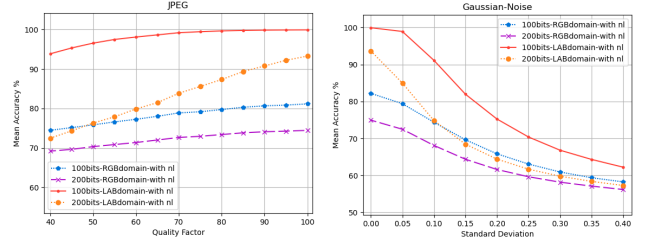


Fig. 3. Robustness performances against JPEG compression (left) and Gaussian noise addition (right), corresponding to the image processing operators included in the noise layer.

Such modification is not enough to affect the watermark, when this is embedded in the *ab* components of the image.

TABLE II
ROBUSTNESS AGAINST IMAGE PROCESSING. FOR BLURRING WE SET THE KERNEL SIZE TO 9×9 AND $\sigma = 5$.

	Blur	Median 5×5	Median 7×7	Contrast Brightness Adj.	HSV Transf.	P&S
Prop Lab, $l = 100$, with nl	99.9%	99.6%	99.3%	97.3%	99.9%	96.3%
RGB, $l = 100$, with nl	80.3%	74.6%	73.6%	82.9%	84.3%	55.4%
Prop Lab, $l = 200$, with nl	88.0%	90.5%	88.3%	87.7%	93.0%	96.6%
RGB, $l = 200$, with nl	73.6%	65.3%	64.9%	71.3%	84.27%	55.5%

C. Model simplification

Finally, we assessed the accuracy when the watermarked GAN is simplified by pruning some of its nodes. The results we got show that with the proposed watermarking colorization scheme the robustness against model simplification is good, and the watermark can be retrieved with pretty large accuracy also from the images generated by the modified models. In particular, the *Lab* model with a payload of 100 bits (res. 200 bits) still achieves an accuracy of 95.1% (res. 85.5%) even when 40% (res. 30%) of the nodes are pruned. On the other hand, for the model working in the RGB domain the accuracy is only 71.7% with 40% pruning in 100 bits case, and 76.8% with 30% pruning in the 200 bits case, demonstrating once again the advantages of embedding and retrieving the watermark in the *ab* domain.

VI. CONCLUDING REMARKS

We have proposed a new method to watermark a GAN network for image colorization. In contrast to existing methods, our method embeds and reads the watermark from the chrominance components of the image, so to match the way colorization networks work. To improve the robustness of the watermark, we introduced a noise layer in the training of both the watermark decoder and the colorization network. The experiments confirm that the working in the CIE-*Lab* domain provides better performance than working in the RGB domain. Although the proposed scheme has been implemented considering a GAN architecture, a similar approach can be extended to watermark colorization networks based on different generative techniques, e.g. language-guided networks or diffusion models. Future work may also focus on the application on the application of the proposed method to other kinds of datasets.

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REFERENCES

- [1] Y. Uchida, Y. Nagai, S. Sakazawa, and S. Satoh, "Embedding watermarks into deep neural networks," in *Proceedings of the 2017 ACM on International Conference on Multimedia Retrieval*, 2017, pp. 269–277.
- [2] M. Barni, F. Pérez-González, and B. Tondi, "Dnn watermarking: four challenges and a funeral," in *Proceedings of the 2021 ACM Workshop on Information Hiding and Multimedia Security*, 2021, pp. 189–196.
- [3] Y. Li, H. Wang, and M. Barni, "A survey of deep neural network watermarking techniques," *Neurocomputing*, vol. 461, pp. 171–193, 2021.
- [4] Y. Li, B. Tondi, and M. Barni, "Spread-transform dither modulation watermarking of deep neural network," *Journal of Information Security and Applications*, vol. 63, p. 103004, 2021.
- [5] N. Yu, V. Skripniuk, S. Abdelnabi, and M. Fritz, "Artificial fingerprinting for generative models: Rooting deepfake attribution in training data," in *2021 IEEE/CVF International Conference on Computer Vision*, 2021, pp. 14 448 – 14 457.
- [6] J. Fei, Z. Xia, B. Tondi, and M. Barni, "Supervised gan watermarking for intellectual property protection," in *2022 IEEE International Workshop on Information Forensics and Security (WIFS)*. IEEE, 2022, pp. 1–6.
- [7] J. Kirchenbauer, J. Geiping, Y. Wen, J. Katz, I. Miers, and T. Goldstein, "A watermark for large language models," in *International Conference on Machine Learning*. PMLR, 2023, pp. 17 061–17 084.
- [8] N. Yu, V. Skripniuk, S. Abdelnabi, and M. Fritz, "Artificial fingerprinting for generative models: Rooting deepfake attribution in training data," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 14 448–14 457.
- [9] P. Fernandez, G. Couairon, H. Jegou, M. Douze, and T. Furon, "The stable signature: Rooting watermarks in latent diffusion models," in *IEEE/CVF International Conference on Computer Vision*, 2023, pp. 22 466–22 477.
- [10] L. Yatziv and G. Sapiro, "Fast image and video colorization using chrominance blending," *IEEE transactions on image processing*, vol. 15, no. 5, pp. 1120–1129, 2006.
- [11] N. Purnekar, L. Abady, B. Tondi, and M. Barni, "Improving the robustness of synthetic images detection by means of print and scan augmentation," in *Proceedings of the 2024 ACM Workshop on Information Hiding and Multimedia Security*, 2024, pp. 65–73.
- [12] H. Wu, G. Liu, Y. Yao, and X. Zhang, "Watermarking neural networks with watermarked images," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 31, no. 7, pp. 2591–2601, 2020.
- [13] M. Sharma, M. Makwana, A. Upadhyay, A. P. Singh, A. Badhwar, A. Trivedi, A. K. Saini, and S. Chaudhury, "Robust image colorization using self attention based progressive generative adversarial network," *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 2188–2196, 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:198906079>
- [14] R. Zhang, P. Isola, and A. A. Efros, "Colorful image colorization," in *Computer Vision – ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part III 14*, 2016, pp. 649–666.
- [15] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," *CVPR*, 2017.
- [16] P. Vitoria, L. Raad, and C. Ballester, "Chromagan: Adversarial picture colorization with semantic class distribution," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2020, pp. 2445–2454.
- [17] C. Saharia, W. Chan, H. Chang, C. Lee, J. Ho, T. Salimans, D. Fleet, and M. Norouzi, "Palette: Image-to-image diffusion models," in *ACM SIGGRAPH 2022 conference proceedings*, 2022, pp. 1–10.
- [18] Z. Chang, S. Weng, P. Zhang, Y. Li, S. Li, and B. Shi, "L-cad: Language-based colorization with any-level descriptions using diffusion priors," 2023.
- [19] Z. Chang, S. Weng, Y. Li, S. Li, and B. Shi, "L-coder: Language-based colorization with color-object decoupling transformer," in *Computer Vision – ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XVIII*. Berlin, Heidelberg: Springer-Verlag, 2022, p. 360–375. [Online]. Available: https://doi.org/10.1007/978-3-031-19797-0_21
- [20] S. Weng, H. Wu, Z. Chang, J. Tang, S. Li, and B. Shi, "L-coder: language-based colorization using color-object decoupled conditions," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, pp. 2677–2684, 06 2022.
- [21] H. Liu, J. Xing, M. Xie, C. Li, and T.-T. Wong, "Improved diffusion-based image colorization via piggybacked models," *arXiv preprint arXiv:2304.11105*, 2023.
- [22] T. Karras, T. Aila, S. Laine, and J. Lehtinen, "Progressive growing of gans for improved quality, stability, and variation," 2018.
- [23] M. Tancik, B. Mildenhall, and R. Ng, "Stegastamp: Invisible hyperlinks in physical photographs," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 2117–2126.
- [24] R. Shin, "Jpeg-resistant adversarial images," 2017. [Online]. Available: <https://api.semanticscholar.org/CorpusID:204804905>